

DIPESH KUMAR

SOFTWARE ENGINEER & TECHNICAL LEAD

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PROFESSIONAL SUMMARY

Result-driven Software Engineer with expertise in full-stack backend development, algorithms, system design, and hardware-software integration. Proficient in Java, C++, Python, and cloud infrastructure with a strong foundation in mathematical optimization, data structures, and high-performance computing systems. Proven track record of architecting scalable applications, optimizing algorithmic performance, and leading cross-functional engineering projects.

TECHNICAL SKILLS

Languages:	Java, C, C++, Python, SQL, Assembly, LaTeX, HTML5, CSS3
Frameworks & Tools:	Spring Boot, Node.js, React, Git, GitHub, Docker, Linux, Unix, IEEE Standards
Engineering & Math:	Data Structures & Algorithms (DSA), System Design, AC Circuit Analysis, Linear Algebra
Databases & Cloud:	PostgreSQL, MySQL, MongoDB, Redis, AWS (S3, EC2), CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Software Engineering Intern - Apex Systems & Technologies May 2025 - Aug 2025

- Architected and deployed microservices using Java (Spring Boot) and PostgreSQL, reducing operational latency by 28% across high-concurrency API endpoints.
- Designed high-efficiency C/C++ memory management modules for real-time telemetry processing, handling 100k+ data packets/sec with zero memory leaks.
- Implemented unit and integration testing pipelines leveraging Docker and GitHub Actions, achieving 94% code coverage and accelerating release cycles by 2 days.
- Collaborated in an Agile team of 6 engineers, participating in bi-weekly sprints, technical design reviews, and automated CI/CD pipeline deployments.

Technical Project Lead - Academic Engineering Labs Dec 2024 - Apr 2025

- Led a 4-member engineering team in designing full-stack simulation software for high-frequency electrical circuit analysis and linear independence vector operations.
- Implemented manual Gaussian elimination and matrix rank solvers in C++ to achieve precise basis determination without external mathematical library overhead.
- Drafted comprehensive IEEE-standard technical documentation and presentation systems using LaTeX Beamer, securing top performance evaluation across academic panels.

KEY PROJECTS

Algorithmic Balance & Recursive Parser Engine Java | Data Structures | Algorithms

- Developed a high-performance Java recursive parser using backtracking to generate and validate complex balanced syntax patterns with $O(2^N)$ space-time optimization.
- Integrated robust dynamic memory management and benchmarked performance against standard Java Collections, reducing space complexity footprint by 18%.

Sustainable Urban Energy & Waste Analytics Dashboard Python | PostgreSQL | Data Visualization

- Engineered an analytical reporting dashboard tracking Waste-to-Energy (WTE) metrics and city pollution indicators across various municipal datasets.
- Automated data ingestion pipelines, structured normalized relational schemas, and generated clean, automated infographic reports for sustainability assessments.

EDUCATION

Bachelor of Technology (B.Tech) in Technical & Electrical Engineering 2023 - Present

Vignan's Foundation for Science, Technology & Research | CGPA: 8.9 / 10.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, AC Circuit Analysis, Linear Algebra & Vector Spaces.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certified Developer - Associate:** Validated skill in building and deploying cloud-based applications on AWS.
- **IEEE Academic Excellence Award:** Recognized for outstanding technical report writing and presentation in IEEE standardized format.
- **Competitive Programming:** Solved 350+ data structure and algorithm problems across LeetCode and HackerRank (Java/C++).